

Peroneal reaction time in treated functional instability of the ankle

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Summary

Peroneal reaction time was measured in two groups of patients with either acute or chronic unilateral instability of the ankle, both before and after treatment. Group A consisted of patients with chronic functional instability who were treated with surgical stabilization while Group B contained patients with acute functional instability who were treated with proprioceptive exercises. Statistical analysis of the electromyographic recordings taken from peroneus longus muscles of these patients showed longer mean peroneal reaction time before conservative or surgical treatment in both groups ($P<0.001$). Group A patients showed no statistical difference in mean peroneal reaction time before and after surgical stabilization; however eight out of 10 patients in this group were satisfied with the surgical outcome of functional instability. Group B patients showed mean peroneal reaction time to have improved ($P<0.001$) following proprioceptive exercises and all of them had satisfactory outcome of treatment. Our study demonstrates that peroneal reaction time is prolonged in patients with acute or chronic unilateral functional instability suggesting disruption of the reflex pathway between ankle and spinal cord. Surgical stabilization improves symptoms of chronic functional instability but no significant change in the peroneal reaction time is demonstrated while proprioceptive exercises improve patients' symptoms of acute functional instability as well as peroneal reaction time. We discuss the possible causes of variations in peroneal reaction time before and after treatment in functional instability of the ankle.

Keywords: functional instability; unstable ankle; peroneal reaction time

Introduction

Injury of the ankle ligaments is one of the most common sports related injuries and among the most

common of all injuries seen by orthopaedic surgeons [1,2]. The vast majority of injuries (85%) are of a single type sprain [3] and an equally high proportion of these involve the lateral ligament complex [4]. Garrick [5] showed that ankle sprains were most common in basketball, with an incidence of 38% in

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men and 45% in women. Football had the next highest frequency with 13.2% of all injuries involving the ankle.

While there is a debate as to the best initial treatment for an acute complete tear of lateral ligament complex of ankle, persistent functional instability of the ankle develops in $\approx 20\%$ of the patients regardless of whether the initial treatment is surgical or conservative [6–11].

Functional instability, a subjective complaint, is the commonest residual disability [12–16] and is described as a sensation of the ankle 'giving way' and a tendency for frequent recurrent spraining. Functional instability is different from mechanical instability, which is abnormally increased mobility of the ankle and indicates a lesion of the passive stabilizers of the ankle, primarily the ligaments. Freeman *et al.* [13] suggested that functional instability of the ankle is secondary to joint deafferentiation and motor incoordination with interference of the reflexes that depend on articular mechanoreceptors. The hypothesis is well supported by studies showing the presence of mechanoreceptors in the ligaments of the human ankles and other joints [17, 18].

Konradsen *et al.* [19] suggested that the loss of afferent input from the torn ligaments may render the ankle susceptible to undetected inversion positions and therefore causing repeated injuries.

Functional instability of the ankle joint is a complex syndrome, in which different mechanical, functional and neuromuscular factors are probably involved [20]. Staples [14] discussed four possible causes of functional instability, which are: (1) peroneal weakness; (2) mechanical instability; (3) tibiofibular sprains and (4) proprioceptive defects.

The measurement of the reflex time of the peroneus muscle group contraction in response to the inversion of the ankle is one way of obtaining information about functional instability. Konradsen and Raven [21] and Karlsson *et al.* [20] measured reaction time from electromyographic (EMG) recordings of the peroneal muscle group by calculating the time interval from the onset of inversion of the ankle to the first electromyographic response, which was called as peroneal reaction time. This time interval was found to be longer in patients with functional instability compared to normal subjects and was described as secondary to abnormal stretch reflexes

through a spinal pathway following afferent nerve fibre injury. They also discounted that any supraspinal or central mechanism was involved in the production of abnormal peroneal reaction time.

The aim of this prospective study was to evaluate the preoperative and postoperative peroneal reaction time in patients with unilateral chronic functional instability of the ankle and compare it to their recovery status in terms of symptoms and function after the operation. It was also of interest to see how peroneal reaction time responded in patients with acute unilateral functional instability following the use of proprioceptive exercises as this appears not to have been previously reported.

Materials and methods

Two groups of 10 patients were selected. Group A included patients with unilateral chronic functional instability of the ankle that were treated surgically. Group B consisted of patients with unilateral acute functional instability of the ankle that were treated with proprioceptive exercises. The experiment was approved by the ethical committee of Liverpool John Moores University. All subjects gave their informed consent to participate in the study.

Group A included eight men and two women, the mean age was 26.5 years (range 16–34 years), seven patients had ankle instability on the right and three on the left side (Table 1). The contralateral ankles were normal and stable in all cases. All of them had history of inversion injury to the ankle of more than 6 months and had received conservative treatment in the form of rest, elevation of the ankle, ice application and dorsiflexion/plantarflexion exercises of the ankle joint once the pain settled. Two of them (patients 3 and 6) received proprioceptive exercises to the injured ankles approximately 3 months after the initial injury to the ankle for the symptoms of ankle giving way but failed to complete the physiotherapy course. All the patients were unable to participate in recreational or sports activities. None of them had any functional instability of the ankle on the injured side in the past. Examination of the bilateral subtalar joints in all patients showed pain-free symmetrical range of movements and the position of the heel on standing and sitting postures revealed no abnormality. Rest of the examination of ankle and foot on both sides was normal. These

Table 1
Details of patients in Group A

Case	Age	Sex	Side	Profession	Mech. of injury	Sports	Pre-op physio	A.D./T.T.*	Outcome	Complications
1	25	M	R	Unemployed	Caught ankle in pot hole	Football	No	Positive	Unsatisfied	Painful around ankle
2	32	F	R	Housewife	Inversion injury, walking	No sports	No	Positive	Satisfied	Nil
3	20	M	L	Joiner	Fell off scaffolding	Boxing	Yes	Positive	Satisfied	Nil
4	16	M	R	Student	Football injury	Football	No	Positive	Satisfied	Nil
5	34	M	R	H.M. Navy	Inversion injury on the ship	Football	No	Positive	Satisfied	Pain in the scar
6	28	M	L	Laundry worker	Squash injury	Squash	Yes	Positive	Satisfied	Nil
7	33	M	R	Unemployed	Fell from platform	Football	No	Positive	Unsatisfied	Nil
8	23	M	R	Student	Inversion injury, walking	Weight lifter	No	Positive	Satisfied	Nil
9	24	M	L	Unemployed	Football injury	Football	No	Positive	Satisfied	Nil
10	30	F	R	Recovery nurse	Twisting injury, walking	No sports	No	Positive	Satisfied	Nil

* A.D. = Anterior Drawer Test; T.T. = Talar Tilt.

Table 2
Details of patients in Group B

Case	Age	Sex	Side	Profession	Mech. of injury	Type of sports	Pre-op physio	Outcome	Return to P.I.L.*
1	24	M	R	School worker	Football injury	Football	Yes	Satisfied	12 weeks
2	28	M	L	Self-employed	Football injury	Football	Yes	Satisfied	10 weeks
3	28	M	R	Unemployed	Inversion injury, walking	Weight trainer	Yes	Satisfied	12 weeks
4	27	M	L	Unemployed	Football injury	Football	Yes	Satisfied	8 weeks
5	25	M	R	Joiner	Fell from height	No sports	Yes	Satisfied	9 weeks
6	24	M	L	Unemployed	Inversion injury, walking	No sports	Yes	Satisfied	10 weeks
7	22	M	L	Student	Football injury	Football	Yes	Satisfied	8 weeks
8	20	M	R	Student	Football injury	Football	Yes	Satisfied	8 weeks
9	23	M	L	Student	Football injury	Football	Yes	Satisfied	10 weeks
10	27	M	R	Self-employed	Football injury	Football	Yes	Satisfied	7 weeks

* P.I.L. = Pre-injury level.

patients were diagnosed to have chronic functional instability on the basis of history of inversion injury to the ankle of more than 6 months, pain over the lateral ligament complex, feeling of ankle giving way, positive anterior drawer test on examination as described by Marder [22] and demonstration of significant talar tilt and anterior translation of talus [23] on stress views of the ankle under general anaesthesia performed by one consultant orthopaedic surgeon. None of the patient's symptoms suggested causes other than lateral ligament injury in the ankle. Special attention was paid to rule out any subtalar joint pathology. Subsequently these patients had lateral ligament reconstruction, which was carried out by modified Chrisman Snook procedure using split peroneus brevis tendon [24]. No attempt was made during surgery to dissect out all the components of the lateral ligament complex of the ankle. Postoperatively all patients had a below knee cast for 6 weeks following which they were referred to physiotherapy department for mobilization exercises to the ankle and subtalar joint. It was noted in the postoperative period that on average all patients had lost 10° of inversion in subtalar joints on the operated side at the end of the rehabilitation programme compared to preoperative assessments but did not complain of pain or walking difficulty on plain or uneven surfaces.

Group B included 10 male patients; the mean age was 24.8 years (range 20–28 years). Right and left ankles were involved equally. Eight patients were active sportsmen and seven of them had inversion injury while playing football (see Table 2). Their contralateral ankles were normal and stable. These patients were diagnosed as having acute lateral ligament injury on the basis of a history of recent inversion injury to the ankle; pain, swelling and tenderness over the lateral ligament complex, pain on active inversion of the ankle and negative radiographs for any fracture in and around the ankle joint. No pathology was noted in the subtalar joint or other joints of the injured foot. No attempt was made to carry out stress views under general anaesthesia. Patients with any other associated injury to the same ankle and foot were excluded from the study along with those who had previous ankle injury to the same side. These patients were treated conservatively with rest, elevation, ice application and compression bandage. Ankle dorsiflexion and

planterflexion exercises started once swelling and pain settled. They were mobilized non-weight bearing to full weight bearing according to the severity of pain. However all of them developed symptoms of recurrent pain and the ankle giving way over the following 6–8 weeks. They were subsequently referred for proprioceptive exercises on a wobble board under the supervision of a senior physiotherapist once diagnosis of acute functional instability was made. The programme for proprioceptive exercises lasted for average 6 weeks.

Electromyographic studies

Electromyographic (EMG) measurements were taken from peroneus longus of the affected and contralateral normal side in both groups before and after the surgical or conservative treatments. The normal side was used as control. Patients were briefed about the testing method and a demonstration was made beforehand by one of the authors to familiarise the procedure to the patients.

A trap door able to tilt 30° in the coronal plane was made to simulate an ankle inversion effect. It measured 20" in width, 17" in length and 8" in height. The middle third part was hinged for tilting and designed to lock. The trap door was connected to two sensors one of which was stimulated when lock of the hinge opened by manually pulling a support slot and consequently tilting the trap door. The other sensor would activate when the door hit the 30° restrainer.

Patients had the hair over the belly of the peroneus longus shaved with a disposable razor after identification of its borders; and a small area on the bony anteromedial surface on the tibia for application of surface electrodes. The skin was cleaned and dried with 70% sterile isopropyl alcohol wipe (Seton Pebbles Ltd, Oldham, UK). EMG signals were recorded using surface electrodes (Nicolet, Nicolet Biomedical Inc., Madison, WI, USA). Two electrodes were centred over the peroneus longus belly 2 cm apart and 3 cm distal to the head of fibula; the third electrode was used as neutral over the tibia.

The subjects would stand with the affected foot on the hinged part and the other foot on another platform of the same height. The intermalleolar distance was set up at 25 cm for normal balance and axis of rotation of the trap door was adjusted just

medial to the sole of the foot. The subjects were instructed to distribute their weight equally on both feet and were unaware of when the trapdoor released; which was carried out once basal activity of EMG was attained as assessed from on line display. The EMG activity was registered using a Biological Telemetry System (MIE Medical Research Limited, Leeds, UK) which consisted of EMG signal amplifier, radio receiver and transmitter; and analogue to digital converter. This apparatus was connected to a computer (Acorn, Cambridge, UK).

The amplified EMG signal from electrodes was sampled at a frequency of 500 Hz. The signal was overlaid with synchronous pulse from the trap door. The peroneal reaction time was recorded as the duration from the time of the trap door signal to the onset of the EMG signal distinguishable above the background noise.

Before treatment group A patients had seven EMG recordings taken while group B patients underwent five EMG recordings for each of the affected and normal sides, by measuring the peroneal reaction time. A mean of the measurements from each patient was taken for analysis. Group A patients had these studies repeated 6 months after their surgery; while group B patients had their measurements repeated at the conclusion of proprioceptive exercises programme and symptomatic recovery.

Comparisons were made using a Student's *t*-test (two tails) and a level of significance was set at $P < 0.05$.

None of the patients faced any difficulty, damage or significant discomfort during and after the testing procedure.

Results

The results of mean peroneal reaction time taken from group A patients comparing injured and normal ankles are shown in Table 3. The mean peroneal reaction time of the injured ankle was significantly longer ($t_9 = 9.57$, $P < 0.001$) in duration compared to the contralateral normal ankles. The large difference between conditions ensured that the test power was high (power >99%) and so the number of subjects used were adequate for the investigation.

Results of peroneal reaction time for group A subjects comparing injured ankles before and after the surgical stabilization are shown in Table 4. The

Table 3

A comparison of the mean peroneal reaction time in group A subjects for the injured and normal ankles

Patients	Mean peroneal reaction time (ms)	
	Injured pre-op	Normal
1	74.85	61.14
2	71.42	60.00
3	69.42	65.71
4	77.00	64.00
5	75.50	61.00
6	72.00	56.28
7	84.82	75.42
8	74.00	64.33
9	72.85	64.57
10	73.42	64.57
Mean	74.53	63.70
SD	4.21	5.01

Table 4

Mean peroneal reaction time in group A subjects before and after surgical stabilization

Patients	Mean peroneal reaction time	
	Injured pre-op	Injured post-op
1	74.85	71.42
2	71.42	72.00
3	69.42	69.60
4	77.00	74.66
5	75.50	73.14
6	72.00	70.58
7	84.82	90.80
8	74.00	72.40
9	72.85	71.14
10	73.42	72.72
Mean	74.53	73.85
SD	4.21	6.21

mean peroneal reaction time was not significantly different before and after the surgery for these patients ($t_9 = 0.82$, $P = 0.433$). Although the surgery had clinically improved the symptoms, this was not reflected in improved reaction time measurements.

The results of peroneal reaction time measurements from group B patients before and after the proprioceptive exercises for injured ankle are shown

Table 5
Mean peroneal reaction time in group B subjects before and after proprioceptive exercises

Patients	Mean peroneal reaction time	
	Injured before	Injured after
1	71.60	62.40
2	83.20	69.20
3	72.40	62.80
4	66.00	71.20
5	79.20	60.40
6	71.60	57.20
7	66.40	63.20
8	75.20	62.40
9	69.60	62.40
10	73.60	61.20
Mean	72.88	63.24
SD	5.33	4.08

in Table 5. The mean reflex time was significantly longer ($t_9 = 4.51$, $P < 0.001$) than the normal side in the acute functional instability as seen in chronic cases; however, following physiotherapy the mean peroneal reaction time had decreased back to normal levels.

Clinically patients 1 and 8 from group A still had pain in the lateral ligament complex region following surgery although the feeling of giving way disappeared. Patient 8 was found to be overweight. The rest of the subjects had satisfactory results from surgery based on the absence of pain and the feeling of giving way, normal walking and participation in light recreational activities. Clinically performed anterior drawer test was negative postoperatively when compared to the normal ankle in all subjects including patients 1 and 8.

Group B patients had a very satisfactory outcome regarding their functional instability and pain. They regained pre-injury activity level at an average 10 weeks following proprioceptive exercises.

Discussion

Freeman [6,13] described a high incidence of functional instability in patients after lateral ligament ruptures regardless of treatment. He hypothesized a proprioceptive defect in the form of de-afferentiation of the lateral ligament and capsule creating functional instability. Freeman *et al.* [25] noted that mechanical

instability of the ankle joints is rarely an initial cause of the symptoms of functional instability. They also postulated that repeated sprains of ankle could cause adhesion formation, which can account for functional instability symptoms. However Karlsson *et al.* [20] showed that functional instability might be a combination of mechanical instability (i.e. increased anterior translation and talar tilt) and proprioceptive deficit (i.e. increased peroneal reaction time during an inversion episode of ankle).

Our results showed that the patients in both groups demonstrated an increased peroneal reaction time in the injured ankle reflecting abnormality in the reflex arc, substantiating the hypothesis of de-afferentiation of the lateral ligament and capsule of the injured ankle joint.

We did not find a statistical difference between the pre- and postoperative peroneal reaction time in group A patients, indicating that peroneus brevis tendon rerouting at the fixed bony points was merely a mechanical means to stabilise the ankle. It did not contribute to the actual reflex arc pathway between afferents of the ankle and motor action of peroneal muscles.

However, peroneal reaction time of the patients treated with wobble board proprioceptive exercises in group B showed a significant decrease in the duration of the reflex arc. This result leads us to following interpretations.

First, if the functional instability is due to de-afferentiation of the ligaments and capsule of the ankle, the proprioceptive exercises should re-establish the interrupted connections of the afferent signals with its efferent component resulting in quicker peroneal response and decreased peroneal reaction time. This hypothesis raises the question of re-establishment of the neuronal pathways, which is very unlikely following physical disruption of the reflex arc as the nervous tissues do not heal with such an accuracy in humans as to establish the exact alignment and function.

Second, the remaining receptors in the injured capsule and ligaments adapt to compensate for the severed reflex connections and fire more rapidly and energetically reducing the peroneal reaction time to pre-injury level. The objection to this idea is then why patients with chronic instability did not improve their peroneal reflex time in the course of time with exaggerated neuronal discharge without any

treatment, while those who did receive proprioceptive exercises showed improvements clinically and electrophysiologically, in a short duration?

Third, proprioceptive exercises made the peroneal muscles more sensitive to inversion signals from ankle and allowed them to respond in a normal way as before the ankle injury rather than showing the characteristic slow reflex response as seen in EMG recordings after the injury to the ankle. It may be, when the afferent signals from the ankle joint are disrupted, the brain and other cerebral higher centres quickly learn to replace this information with that from the mechanoreceptors of the peroneal muscles and tendons for the precise judgement of position of the ankle once the peroneal muscle complex has been re-educated with proprioceptive exercises. In other words functional training preserves the nutrition and enhances the co-ordination of the peroneal musculature in stabilizing the ankle joint.

The view that the peroneal muscles are re-educated is strengthened by the fact that eight out of ten patients with chronic instability in group A never received early proprioceptive exercises to train peroneal muscles to react effectively and efficiently in preventing symptoms of the functional instability. While all the patients in group B received prompt and proper exercises, once the diagnosis of functional instability was made, to stabilise their ankles by involving the brain and higher centres in the training process well supervised by trained physiotherapy staff.

The third interpretation also suggests a further study in which proprioceptive exercises are performed in late diagnosed cases of functional instability of the ankle with a recording of the peroneal reaction time before and after the physiotherapy. This will help to understand whether muscle re-education can occur in chronic functional instability, as we do know if proprioceptive exercises do improve peroneal reaction time in chronic functional instability.

Conclusions

Our study confirms that peroneal reaction time was prolonged in patients with both acute and chronic functional instability of the ankle as compared to the normal side. While surgical stabilization improved

patient's symptoms, no significant change in peroneal reaction time was observed in chronic cases.

Patients with acute ankle sprains and poor peroneal reaction time at 6–8 weeks can have significant improvement clinically and measurable improvements in peroneal reaction time with proprioceptive exercises of the ankle.

We must stress that two groups of subjects were not compared to each other in this study because of many variables, which could not be controlled.

There is need for a further study to be carried out to record peroneal reaction time in chronic unstable ankles following proprioceptive exercises to identify any changes which may accompany the treatment.

Statement of interest

No benefits in any form have been received or will be received from a commercial party related directly or indirectly to the subject of this article.

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